Final Exam - Applied Abstract Algebra

June 9th, 2026

(40 points) Let Q be a quiver:

$$\begin{tikzcd}
 & & & 4 \\
1 & 2 \arrow[l] \arrow[r] & 3 \arrow[ur] & \\
 & & & 5 \arrow[ul]
\end{tikzcd}$$

a). Construct $P(2)$ and $I(3)$ and using projective and injective resolution calculate $\tau(P(2))$ and $\tau^{-1}(I(3))$, where τ denotes the Auslander-Reiten translation.

b). Construct a triangulation of punctured polygon corresponding to Q and find which arcs with respect to this triangulation correspond to $P(2)$, $I(3)$, $\tau(P(2))$, $\tau^{-1}(I(3))$ and to indecomposable representation of dimension vector $(1, 2, 2, 1, 1)$.

Choose up to three problems from this list to submit.

1. *(20 points)* Prove that P is projective if and only if $\mathrm{Ext}^1(P, N) = 0$ for all representations N.

2. *(20 points)* Let $M \in rep\, Q$ and

$$0 \to P_1 \to P_0 \to M \to 0$$

be standard projective resolution. Show that

$$0 \to D(M) \to D(P_0) \to D(P_1) \to 0$$

is the injective resolution for $D(M) \in rep\, Q^{op}$. Show that duality functor D takes a minimal projective resolution for M to a injective envelope for $D(M)$.

3. *(20 points)* Let $g : M \to N$ be a surjective morphism between representations of Q, and let $P(i)$ be the projective representation at vertex i. Show that the map

$$g_* : \mathrm{Hom}(P(i), M) \to \mathrm{Hom}(P(i), N)$$

is surjective.

4. *(20 points)* Show that for any Q the representations $I(i)$, $i \in Q_0$ are indecomposable.

5. *(20 points)* Show that the smallest wild tree with valence at most 3 at each vertex has 7 vertices.